

同济大学课程考核试卷 (A 卷) (

2006 2007 : H1 ' :

Q N 1 - 1R E' u - 1R E'

A , 030238 A E' " (|A`) : 5 A ' 5 " 5 A '
!8 KE ~ 5(A) ^ 4 5 A') ^G; 5()A 'K

H(_____ : _____ SE _____ k _____

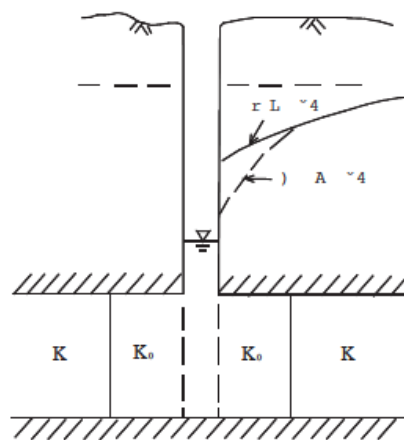
. ?ON (20%~"! £ON0.5%)

1. " | o : -孔隙 "裂隙 、岩溶溶隙 E / | ? _
, X/¥ : ~"E 具有空隙的岩土体 /~ (B ~
2. #°# ~" 4 流线 、 4 等水头线 4 , X5% /~ # 5%J ~ ~"
各向同性 B # 5% !7 x5% ~
3. #°# E/ | ?U2 流量 ^ 流速 ^ 压力 、 水头 1 1 ~
4. # ° E a) [岩土介质的 B B ~ " 5 # ' , X B B . G ~ " # ° E a) [, X
cm² (E D(Darcy) ~ " G£4 L² ~
5. B M2 B } 岩土性质 (透水性) 随空间坐标 , X G2~ , X ~"
B ' B } 岩土性质 (透水性) 随水流方向 G2~ , X ~
6. 9 # ° > 9 4 > , X " # K " " " ~ " ?U' , £ £ " " ! E, 1 ° ~"
" # 高水位河渠 { ~ V p 9 # ° > 9 4 > W > 0 ° ~ " # z ~ 4 , X
6 (椭圆曲线 ~ ' W < 0 ° ~ " 双曲线 ~ ' W = 0 ° ~ " 抛物线 ~
- 7 ~ ° " S _ " i 0 n E | , 井轴的径向直线 4:1 " M6 以井轴为共轴的圆柱面 ~ M6# G£处处相等 ~
- 8 ~ ° C ^ # . 2 ~ C ^ A B ~ " E " 越大 n " z J # E a 越小 ~ " C #
G £ 越小 ~
- 9 ~ ° " ~ @ a i , X ^ ! £ M 6 , X " L ! E < £ ? _
由小增大 ~ " 1/u=1 ° E 最大 ~ " 5 由大变小 ~ " C 等速下降 ~
- 10 ~ ° " E " 空隙 、 固体颗粒 4 B, X M6 / # ° E a # E 整个过

水断面 上的 平均 速度，而实际速度是 $\frac{K_r}{\mu} \frac{\partial h}{\partial r}$, X G E z ~

11. BE \$, < " Efl" F... !, X E E~" H i (=) ~" =
 £) ' =£ c = % n) 20 ~

^ ± N1~12%~"! £N1 t1 (! 7. B+2% ^ t1 (J A^~2% ^ E t1 (0%~"! 7. B, X
 1 (N14 m ~"J A^, X m V ~-
 1~ ° 1 / , XL!: \$^ k ~4 6 (>< ~" , X# K~K (Q V)



1

2. ·A " " , X- E z V) < E~"A,, " , X#°E^a) [(E < , X ~

3. B " ~" & , X#°E^a2~ DF (,) ~

4~°E ?S @^a E °K~ < G£~" ,,E ?S @^a E (0V) n# ~

5~° ? U 4 > n E + " , ^ j .B"n " iL E¥, X" 5 E1'u#°

E^a2~ D ' " G£, X B^a V())~

6~° M 2 0 n i # ~" " # M 6 # G £ E # r B~" + b'
 B, X4 (~V)

^1TE~N1~8~ [N1A,* 8~ [t1 (~28%~-

1~°Give a brief introduction to curve-match method that can be used for determining
 aquifer parameters for an unsteady radial flow in a confined aquifer. ~10%~-

Solution:

1. Theory ~4%~-

$$\text{As: } s = \frac{Q}{4\pi T} W(u) \quad u = \frac{r^2 \mu^*}{4Tt}$$

So:

$$\lg s = \lg W(u) + \lg \frac{Q}{4\pi T}$$

$$\lg \frac{t}{r^2} = \lg \frac{1}{u} + \lg \frac{\mu^*}{4T}$$

The relation between $W(u)$ and $\frac{1}{u}$ similar to that between s and t/r^2 .

2 Steps: (6%)

1. Plot the time-drawdown data on log-log graph paper .
2. Superimpose this plot on the type curve sheet of the same size and scale as the time-drawdown plot , so that the plotted points match the type curve . The axes of both graphs must be kept parallel.
3. Select a match point ,which can be any point in the overlap area of the curve sheets .
4. Determining the parameters:

$$T = \frac{Q}{4\pi[s]} [W(u)] \quad \mu^* = \frac{4T}{\left[\frac{1}{u}\right]} [t]$$

2. 镜像法中，虚拟井的特征。(8% 各 2%)

- 1) 虚井实井的位置以边界为对称;
- 2) 虚井的流量与实井相等;
- 3) 虚井的性质取决于边界的性质: 定水头补给边界, 虚、实井性质相反;
隔水边界: 虚、实井性质相同;
- 4) 虚、实井工作时间相同。

3. 图 2 为均质、各向同性的土坝, 水流在土坝中为剖面非稳定二维流, 试写出渗流区的数学模型。

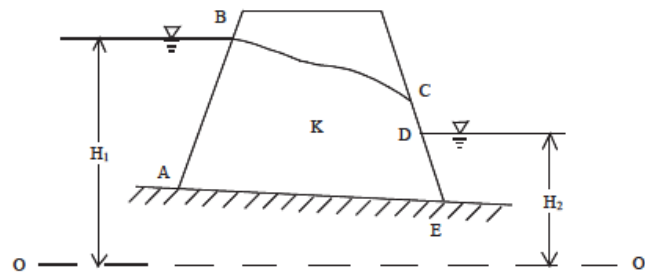


图 2

解:

$$\frac{\partial}{\partial x} \left(h \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial z} \left(h \frac{\partial h}{\partial z} \right) = \frac{\mu}{K} \frac{\partial h}{\partial t} \sim 5\%$$

$$h(x, z, t) \Big|_{t=0} = h_0(x, z)$$

$$AB: h=H_1 \sim 1\% \text{ each}$$

$$BC: \frac{\partial h}{\partial n} = 0$$

$$CD: h=z$$

$$DE: h=H_2$$

$$AE: \frac{\partial h}{\partial n} = 0$$

if $\frac{\partial h}{\partial n} \sim 8\%$ $\frac{\partial h}{\partial n} \sim 8\%$ $\frac{\partial h}{\partial n} \sim 40\%$

1. A well is located in a confined aquifer with a conductivity of 14.9m/d and a storativity of 0.0051. The aquifer is 20.1 m thick and is pumped at a rate of 2725m³/d.

What is the drawdown at a distance of 7.0m from the well after 1 day of pumping?

($w(1.5 \times 10^{-4})=8.23$, $w(2.0 \times 10^{-4})=7.94$, $w(2.5 \times 10^{-4})=7.72$) .(10%)

Solution:

$$T=Kb=14.9\text{m/d} \times 20.1\text{m}=299\text{m}^2/\text{d}$$

$$u = \frac{r^2 S}{4Tt} = \frac{(7.0\text{m})^2 \times 0.0051}{4 \times 299\text{m}^2/\text{d} \times 1\text{d}} = 0.0021u$$

$$\text{As : } w(2.0 \times 10^{-4})=7.94$$

$$\text{So: } h_0 - h = \frac{Q}{4\pi T} W(u) = 5.7\text{m}$$



K'Nl ^ „G * Au1k "'^" J B#°# E+4` B^S2%?~ ~

F... ✓

$$) \quad " \quad F \dots \quad q = k_2 \frac{M^2 - h_2^2}{2(l-x)}$$

$$E^{\sim} \models \langle E^{\sim}, X \# G \rangle, 1 \sim$$

" $k^* x = 1166.7m$ ~8% "

$$(2) \quad) \quad \#G\mathbb{L}^\vee$$

$$q = k_2 \frac{M^2 - h_2^2}{2(l-x)} = 10 * \frac{20^2 - 10^2}{2(2000 - 1166.7)} = 1.8 m^3 / d \quad (5\%)$$

3. In Fig 4, two wells fully penetrate a confined aquifer, $K=5.25 \times 10^{-4} \text{ m/s}$, $M=35.0 \text{ m}$, $r_w=0.2 \text{ m}$, $R=100 \text{ m}$, $a=29.5 \text{ m}$, $b=50.5 \text{ m}$, $c=25 \text{ m}$. When a drawdown of 10m in well 2 and a steady state reaches, find Q . (15%)

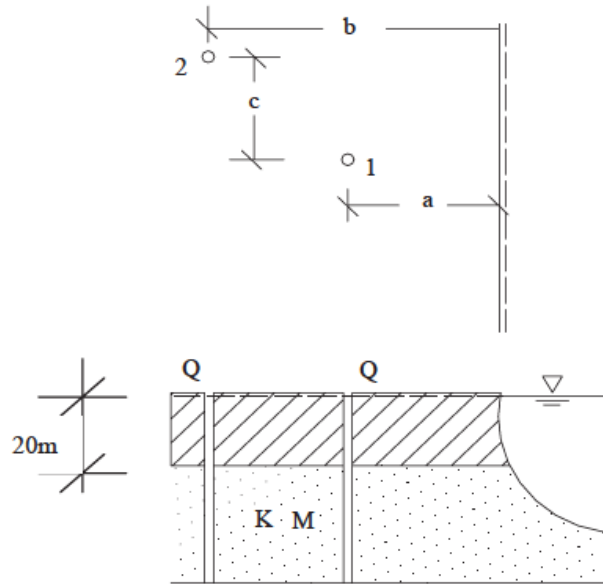


Fig 4

Solution:

As: $a=29.5 \text{ m} < R/2$; $b=50.5 > R/2$ (5%)

So, the boundary only influence the well 1, and at the same time, as the boundary is a recharging boundary, so the image well of well 1 is a recharge well.

So, when well 1, 2 work at the same time, drawdown in well 2 can be calculated as:

$$s = \frac{Q}{2\pi T} \ln \frac{R}{r_w} + \frac{Q}{2\pi T} \ln \frac{R}{r_1} - \frac{Q}{2\pi T} \ln \frac{R}{r_2} = \frac{Q}{2\pi T} \ln \frac{Rr_2}{r_w r_1} \quad (8\%)$$

When the drawdown in well 2 reaches 10m, flux:

$$Q = 2\pi T \cdot s \frac{1}{\ln \frac{Rr_2}{r_w r_1}} = 0.161 (m^3 / s) \quad (2\%)$$